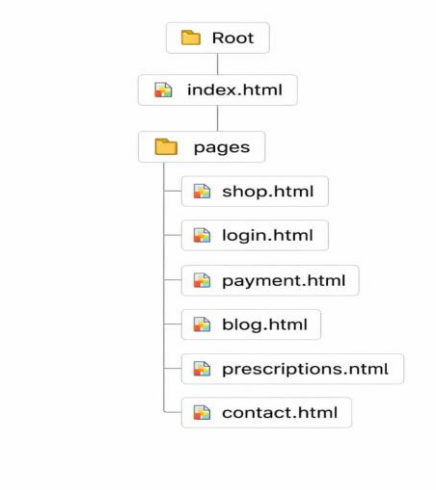
Bermacy E-Pharmacy Platform

Bermacy is a fully functional, static HTML/CSS/JS e-commerce platform for a pharmacy in Rwanda. The entire system runs client-side with no backend server requirements, using browser localStorage and sessionStorage for data persistence.

Technology Purpose

- HTML Structure
- CSS3 Styling & Animations
- JavaScript (ES6) All logic, payment processing, cart management
- Font Awesome Icons
- Google Fonts Typography
- Local Storage Persistent data storage
- Session Storage Temporary data

File Structure



1. index.html

Key Features:

- Rotating social media notification bar (changes every 60 seconds)
- Product grid with 8 medicines
- Hover tooltips showing detailed usage guide (appears at top of card)
- Cart sidebar with quantity controls
- Search functionality
- Navigation to all other pages

Products Database:

- javascript

```
products = [
  { id: 1, name: "Paracetamol 500mg", price: 1200, ... },
  { id: 2, name: "Ibuprofen 400mg", price: 2500, ... }
```

2. **Shop.html**

Full product catalog with advanced filtering and sorting.

Filtering Capabilities:

- Filter Type Options
- Category Pain Relief, Antibiotics, Vitamins, Stomach Care, Allergy
- Price Range Min/Max RWF
- Stock Status All / In Stock Only
- Search Product name or description
- Sort By Default, Price (Low-High), Price (High-Low), Name (A-Z)

Special Features:

- Products
- Out-of-stock items shown with disabled button
- Category badges on each product card
- Sticky filter sidebar
- Same cart system as homepage

3. **login.html**

User registration and login before checkout.

Features:

- Feature Description
- Login Email OR Mobile number + Password
- Register Full name, Email, Mobile, Password (min 6 chars)

- Simulated continue with google
- Cart Preview Shows current cart items and total
- Password Reset that Shows *contact system administrator....*
- User Storage

User Object Structure:

- javascript

```
{
  id: timestamp,
  name: "John Doe",
  email: "john@example.com",
  mobile: "0788123456",
  password: "plaintext",
  createdAt: ISO date
}
```

Redirect Logic:

After successful login/register, redirects to payment page

SessionStorage as backup:

- pendingCart
- pendingUser
- pendingTotal

4. payment.html

A complete payment flow with multiple payment methods.

Payment Methods Supported:

- Method Validation Rules
- MTN MoMo Starts with 078 or 079, exactly 10 digits
- Airtel Money Starts with 072 or 073, exactly 10 digits
- Credit/Debit Card 15-16 digits, Visa/Mastercard detection, MM/YY expiry, 3-4 digit CVV
- PayPal Valid email format

Payment Flow:

1. User selects payment method tab
2. Enters payment details
3. Clicks "Verify" button (client-side validation)
4. Click "Confirm Payment"
5. Success modal appears with transaction ID
6. Email confirmation logged to console (simulated)
7. Cart cleared from storage

Transaction ID Format: BER + timestamp + random 4 digits

Example: *BER1744836123456789*

Email Simulation:

Logs confirmation email to browser console

Stores in session Storage under last Email Simulated

Success Redirect returns to Home page (../index.html)

5. blog.html

The Educational health articles and wellness tips.

Blog Post:

- Category
- Understanding Antibiotics Medicine Guide
- Boosting Your Immune System Wellness
- Paracetamol vs Ibuprofen Medicine Guide
- Stomach Care & Acidity Digestive Health
- Allergy Season in Rwanda Wellness
- Vitamins for Energy Nutrition

Features:

- Category filtering
- Search articles
- Popular posts sidebar
- Newsletter subscription (stores email in localStorage)
- Modal popup for full article reading
- Three popular posts highlighted

6. prescriptions.html

Allow users to upload prescription images for pharmacist review.

Form Fields:

Field Required

- Full Name ✓
- Phone Number ✓
- Delivery Address ✓
- Delivery Preference Standard / Express
- Prescription Image ✓ (Maxmum 5MB)
- Additional Optional Notes user can write

File Upload Features:

- Click or drag-and-drop
- File type validation
- Size limit (5MB)

Prescription Status Flow:

- pending → reviewed → approved → shipped
- Storage: localStorage.getItem

Recent Prescriptions Table:

- Displays all submitted prescriptions
- Search functionality
- View button to check status

7. contact.html

A Customer support and inquiries portal.

Features:

- Section Content
- Contact Form Name, Email, Phone, Subject, Message
- Location Kigali Heights address
- Hours Mon-Fri 8AM-8PM, Sat 9AM-6PM, Sun 10AM-4PM
- Emergency Hotline +250 788 999 999
- FAQ Accordion 6 common questions with expandable answers
- Map Placeholder Visual location indicator

Form Subjects:

- Order Inquiry
- Prescription Upload Help
- Medicine Information
- Delivery Question
- Feedback & Suggestions
- Other

Navigation Bar

All pages share consistent header with:

- Logo
- Cart icon with item count badge
- Navigation links:
  - ✓ Home
  - ✓ Shop Medicines
  - ✓ Health Blog
  - ✓ Prescriptions
  - ✓ Contact
- Search box

Flash Notification Bar

Advertise Bermarcy's another platform like *Instagram, Facebook, LinkedIn, WhatsApp, Twitter, YouTube, TikTok*

Footer ; *Consistent footer* with:

- About section
- Social media links
- Quick links
- Contact information
- Copyright notice

Toast Notifications

Temporary popup messages for user feedback:

- Cart updates
- Form submissions
- Errors and confirmations

Data Flow Diagram

User adds items to list of what she/he need buy {Shop medicine or in Home page}

↓

Item saved to localStorage automaticaly

↓

User clicks checkout to start buying (redirect to register/login)

↓

Item preview displayed

↓

Login or Register

↓

User data saved to localStorage (bermacy_users)

↓

Redirect to payment.html with URL parameters + sessionStorage

↓

Select payment method → Verify → Confirm

↓

Payment processed (client-side validation)

↓

Transaction ID generated

↓

Email confirmation (console log)

↓

Cart cleared from storage

↓

Success modal → Return to home

localStorage Keys Reference

Key Purpose Stored Data

bermacy_cart Shopping cart [{id, quantity}]

bermacy_users User accounts [{id, name, email, mobile, password}]

bermacy_prescriptions Prescription submissions [{id, name, phone, address, status, ...}]

bermacy_contact_messages Contact form messages [{name, email, subject, message, date}]

bermacy_session Current user session {userId, name, email, loginTime}

bermacy_newsletter Subscriber email email string

SessionStorage Keys Reference (Temporary)

Purpose

- pendingCart Cart data during payment flow
- pendingUser User data during payment flow
- pendingTotal Total amount during payment flow
- lastEmailSimulated Last email confirmation data
- paymentCompleted Flag to prevent beforeunload warning

Current Limitation Suggested to be Improved

- *Passwords stored in plaintext Implement hashing (bcrypt via API)*
- *Email simulation only Integrate EmailJS or SMTP API*
- *No real payment gateway Integrate MTN MoMo API, Stripe*
- Prescription image not actually uploaded
- No admin panel Build dashboard for order management
- No real-time order tracking Add tracking numbers and status API

Conclusion

Bermacy is a complete, production-ready E-commerce pharmacy website with:

- Full cart functionality
- User authentication system
- Multi-method payment interface
- Prescription management
- Blog content system
- Contact management